

MD ROBIUL ISLAM

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[Academic Portfolio](#) | [LinkedIn](#) | [Google Scholar](#) | [ResearchGate](#)

RESEARCH INTERESTS

My research interests cover the interdisciplinary application of machine learning (ML), deep learning (DL), reinforcement learning (RL), large language models (LLMs), and multi-agent systems (MARL) in autonomous systems, cyber-physical systems, IoT and 5G/6G wireless networks, network communications, power systems and smart grid, AI-Driven Energy Optimization and agentic AI frameworks. I am actively seeking a PhD position where I can contribute to cutting-edge research at the intersection of artificial intelligence and real-world intelligent systems.

EDUCATION

MS in Computer and Information Systems Engineering (ECE Dept.)

Sep 2024 – Present

Tennessee State University (TSU), Nashville, TN

- Thesis: Optimization of 5G Network Slicing via Deep Reinforcement Learning — Comparative Study of PPO, DDPG, and SAC Algorithms
- Key Courses: Intelligent Systems & ML, Cyber-Physical Systems Modeling & Simulation, Computer Communication & Networking, Modern Signal Processing, Software Systems Design, Probability & Risk Analysis, Systems Engineering

BSc in Electrical and Electronic Engineering

May 2014 – Nov 2019

Pabna University of Science and Technology (PUST), Bangladesh

- Thesis: Fault Analysis of Transmission Lines for Power Systems Using MATLAB Simulation — Developed a MATLAB Simulink model for transmission line fault analysis; achieved 3–5% reduction in fault voltage and fault current, demonstrating effective protection system design
- Key Courses: Electrical Circuits, Power systems engineering, Electrical power transmission and distribution, Artificial Intelligence, Digital Image Processing, Digital Signal Processing, Network Communication Engineering, Digital Electronics

RESEARCH EXPERIENCE

Graduate Research Assistant

Fall 2024 – Present

Dept. of Electrical and Computer Engineering, Tennessee State University | Advisor: Dr. Sagnika Ghosh

- Designed and implemented a deep RL-based 5G network slicing optimization framework comparing PPO, DDPG, and SAC algorithms for dynamic resource allocation in household IoT environments
- Constructed MATLAB-simulated 5G dataset with SNR, Path Loss, and BER parameters; PPO outperformed DDPG and SAC in network reliability and convergence speed
- Quantified 10–15% improvement in network efficiency under dynamic conditions; manuscript submitted and under peer review
- Applied multi-objective optimization and adaptive RL policy design, developing transferable expertise in intelligent decision-making under constrained, real-world conditions

Undergraduate Research Assistant

2019 – 2020

Dept. of Electrical and Electronic Engineering, Pabna University of Science and Technology, Bangladesh | Advisor: Dr. Md Feroz Ali

- Developed MATLAB Simulink fault analysis model for power transmission lines; achieved 3–5% reduction in fault voltage and fault current
- Simulated fault scenarios across multiple network topologies to evaluate system resilience and protection relay performance

RESEARCH PUBLICATIONS

Journal Articles

- [1] Md Robiul Islam, S. Ghosh, "Optimization of 5G Network Slicing Through Reinforcement Learning: A Comparative Study of PPO, DDPG, and SAC Algorithms" MS Thesis Work, submitted and under review, 2026.
- [2] S. Debnath, Md. Buiya, Md. Reza, Md Robiul Islam, et al., "AI-Driven Cybersecurity for Renewable Energy Systems: Detecting Anomalies with Energy-Integrated Defense Data," International Journal of Applied Mathematics, vol. 38, no. 5s, pp. 1002–1027, Aug. 2025. DOI: <https://doi.org/10.12732/ijam.v38i5s.367>

- [3] S. Debnath, Md Robiul Islam, et al., "Hybrid Multi-Scale Deep Learning Enhanced Electricity Load Forecasting Using Attention-Based CNN-LSTM Model" — IEEE Access, 2025. DOI: <https://doi.org/10.1109/ACCESS.2026.3656545>

Peer-Reviewed Conference Papers

- [4] S. Debnath, Md Robiul Islam, et al., "AI-Driven Hybrid Deep Learning Framework for Short-Term Renewable Energy Forecasting under Extreme Weather Events," IEEE ICECIE-2025. DOI: [10.1109/ICECIE66637.2025.11363804](https://doi.org/10.1109/ICECIE66637.2025.11363804)
- [5] S. Debnath, U. Basu, Md Robiul Islam, et al., "Extreme Weather Grid Load Forecasting using Weather-Informed LSTM and Transformer Models," IEEE NAPS, 2025. DOI: [10.1109/NAPS66256.2025.11272315](https://doi.org/10.1109/NAPS66256.2025.11272315)
- [6] J. Hassan, Md Robiul Islam, et al., "Comparative Analysis of ML Models for Multi-Class Activity Recognition Using EEG and Motion Sensor Data," IEEE QPAIN, 2025. DOI: [10.1109/QPAIN66474.2025.11171711](https://doi.org/10.1109/QPAIN66474.2025.11171711)
- [7] J. Hassan, Md Robiul Islam, et al., "EEG-Based Workload Estimation and Classification Using Discrete Wavelet Transform and Statistical Features," IEEE QPAIN, 2025. DOI: [10.1109/QPAIN66474.2025.11171964](https://doi.org/10.1109/QPAIN66474.2025.11171964)
- [8] Md Khan, J. Hasan, Md Robiul Islam, et al., "Optimizing Pomegranate Disease Detection: A Comparative Analysis of Deep Learning Models," IEEE QPAIN, 2025. DOI: [10.1109/QPAIN66474.2025.11171840](https://doi.org/10.1109/QPAIN66474.2025.11171840)
- [9] Md. O. Khan, J. Hasan, Md Robiul Islam, et al., "Design, Structural Analysis, and Challenges of High-Voltage Underground Cables in Sindh, Pakistan," IEEE SPICSCON, Nov. 2025. DOI: <https://doi.org/10.1109/SPICSCON69221.2025.11504100>
- [10] Md. O. Khan, J. Hasan, Md Robiul Islam, et al., "A Wavelet-Statistical Method for EEG Signal Classification of Mental Stress in Cognitive Tasks," IEEE SPICSCON, Nov. 2025. DOI: <https://doi.org/10.1109/SPICSCON69221.2025.11503956>

TECHNICAL SKILLS

Programming: Python, MATLAB/Simulink, C, C++

Reinforcement Learning: PPO, DDPG, SAC, DQN, TD3, Q-Learning, MARL, Policy Gradient — Stable-Baselines3, OpenAI Gym

Deep Learning & ML: PyTorch, TensorFlow, Keras, Scikit-learn, YOLO, ResNet, CNN, LSTM, RNN, GAN, Autoencoders

LLMs & NLP: Transformers, BERT, GPT-2/3, T5, LangChain, OpenAI Codex — MCP Server (OAuth 2.0, RBAC, TLS/SSL)

Optimization: Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Simulated Annealing (SA)

Cloud & DevOps: AWS, Google Cloud Platform, Microsoft Azure, Git/GitHub, High Performance GPU Computing

Visualization & BI: Matplotlib, Seaborn, Plotly, Power BI, Tableau, Google Data Studio

KEY PROJECTS

MCP Server — Secure Middleware for LLM Integration

2025

- Built secure Model Context Protocol (MCP) server enabling standardized LLM–enterprise communication with OAuth 2.0 authentication, RBAC, TLS/SSL encryption, and real-time REST/WebSocket integrations for SQL databases and collaboration tools

Cyber-Physical Systems — Car Dynamics Simulation with Road Grade Disturbance

2025

- Simulated vehicle dynamics on flat and graded roads (4.5° and 9°) using Newton's second law and Trapezoidal numerical method; visualized position and velocity outputs to inform vehicle control system design

Image Restoration Using Spatial and Frequency Domain Filtering

2026

- Analyzed degradation types (Salt & Pepper, Gaussian, Speckle, Periodic Noise) and applied Median and Wiener filters across spatial and frequency domains to restore image quality and minimize noise artifacts

STANDARDIZED TESTS

GRE: Overall: 307 | Quantitative: 159 | Verbal: 148 | AWA: 3.5

Valid: March 2021 – March 2026

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification — Stanford University / Andrew Ng (Coursera)
- Unsupervised Learning, Recommenders, Reinforcement Learning — Stanford University / Andrew Ng (Coursera)
- Advanced Learning Algorithms — Stanford University / Andrew Ng (Coursera)
- Python for Data Science, AI & Development — IBM (Coursera)
- Data Structures and Algorithms Specialization — University of California San Diego, (Coursera)

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern — Join Venture AI, Bangladesh (BD) **March 2023 – Dec 2023**

- Designed and implemented machine learning pipelines for data cleaning, feature extraction, model training, validation, and result interpretation using Python-based AI/ML tools.
- Applied classical ML algorithms and deep learning models to classification, prediction, and pattern-recognition problems, with emphasis on model comparison and generalization performance.
- Conducted experimental evaluation using accuracy, precision, recall, F1-score, confusion matrix, and cross-validation to analyze model reliability under real-world data conditions.

Power System Engineer Intern — Raj Lanka 52.2 MW Power Company Limited, BD *Nov 2018 – May 2019*

- Tested electrical components for resistance defects; supported design and construction of power plant and transmission systems
- Collaborated with senior engineers to troubleshoot and validate new electrical products and distribution infrastructure

PROFESSIONAL AFFILIATIONS & LEADERSHIP

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| • Associate Member, IEEE — Pabna University of Science and Technology Student Branch | 2018–2020 |
| • Secretary, South Asian Student Association (SASA), Tennessee State University | 2024–2025 |
| • Organizing Secretary, Jonaki Voluntary Organization, PUST | 2016–2019 |
| • Math Olympiad Runner-up — District Level | 2011 |

AWARDS & SCHOLARSHIPS

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| • Prime Bank Foundation Meritorious Scholarship — Higher Secondary School Certificate | 2013–2019 |
| • Dutch-Bangla Bank Scholarship — Secondary School Certificate Level | 2010–2012 |
| • Junior School Government Scholarship — Talent Pool Category | 2008–2010 |

REFERENCES

Dr. Sagnika Ghosh: Associate Professor, ECE Dept., Tennessee State University | sghosh1@tnstate.edu | +1 615-963-5362

Dr. Kamrul Hasan: Assistant Professor, ECE Dept., Tennessee State University | mhasan1@tnstate.edu | +1 615-963-7539

Dr. Md Feroz Ali: Associate Professor, EEE Dept., Pabna University of Science and Technology | feroz.ali.eee@pust.ac.bd | +8801724909432